

TalentLens AI

Candidate Ranking System

An AI-Powered Approach to Smarter Hiring

1. Problem Statement

Recruiters review hundreds of candidate profiles and often miss highly relevant candidates for specialized AI/ML roles. Traditional keyword-based filtering overlooks candidates with deep technical depth, non-obvious title matches, and diverse career paths that map to the job requirements.

The Challenge: How can we rank candidates the way a great recruiter would—by understanding who truly fits the role, not just matching keywords?

2. Solution Approach

TalentLens AI uses a **hybrid semantic + rule-based ranking system** that combines traditional talent signals with modern AI embeddings to evaluate candidate fit holistically.

Hybrid Ranking Framework

Component	Weight	Purpose
Rule-Based Scoring	40%	Title, experience, skills, career history, recruiter signals
Semantic Embeddings	60%	Deep understanding of job description and candidate profile alignment

3. Scoring Components

A. Rule-Based Signals (40% weight):

- **Job Title Matching:** AI-aligned roles (+40), disqualifying roles (-60)
- **Experience Range:** 5-9 years target (+25), 4-11 years acceptable (+15)
- **Technical Skills:** Embeddings, retrieval, ranking, vector databases (+5 per match)
- **Career History:** Search engineer (+30), Recommendation systems (+35), ML roles (+25)
- **Recruiter Signals:** Response rate, open-to-work status, search activity

B. Semantic Matching (60% weight):

- Uses Sentence-Transformer embeddings (all-MiniLM-L6-v2 model)
- Embeds job description and candidate profiles into vector space
- Calculates cosine similarity for semantic relevance
- Captures nuanced role requirements beyond keyword matching

4. Ranking Algorithm

For each candidate:

1. Rule-Based Score (S_{rule}):

- Evaluate title, experience, skills, history, and recruiter signals
- Sum weighted scores across all matching criteria

2. Semantic Score ($S_{semantic}$):

- Embed job description: $\text{embedding}(\text{job_description})$
- Embed candidate profile: $\text{embedding}(\text{profile} + \text{skills} + \text{history})$
- Calculate similarity: $\text{cos_similarity}(\text{jd_embed}, \text{candidate_embed}) \times 100$

3. Final Hybrid Score:

- $\text{Final Score} = (S_{rule} \times 0.4) + (S_{semantic} \times 0.6)$

4. Ranking:

- Sort candidates by Final Score in descending order
- Output top 100 ranked candidates to submission.csv

5. Data Pipeline

Input: Candidate data (candidates.jsonl) + Job description (job_description.docx)

Processing: Python ranking engine with embeddings and hybrid scoring

Output: Ranked candidates (submission.csv) with scores

Visualization: React dashboard for shortlist review and search

6. Output Format

The submission.csv file contains ranked candidates in CSV format:

candidate_id	rank	score
CAND_0018499	1	539.00
CAND_0092278	2	502.00
CAND_0033861	3	474.00
...	...	...

7. Key Advantages

Over Traditional Keyword Matching:

- ✓ Understands job requirements holistically, not just keywords
- ✓ Captures relevant experience even with non-standard titles
- ✓ Combines structured rules with semantic understanding
- ✓ Leverages recruiter signals (response rate, availability)
- ✓ Prioritizes production experience and demonstrated impact
- ✓ Transparent scoring (rule-based + semantic) for insights

8. How to Run

Backend (Generate Ranked Output):

```
cd ai-candidate-ranker/Backend/src  
python rank_candidates.py
```

Copy Output to Frontend:

```
cp output/submission.csv ai-candidate-ranker/Frontend/public/submission.csv
```

Start Frontend Dashboard:

```
cd c:\Users\acer\OneDrive\Desktop\AI_Candidate_Ranker\ai-candidate-ranker  
npm run preview
```

View Results:

Open the displayed localhost URL in your browser to explore ranked candidates.

9. Summary

TalentLens AI combines the best of both worlds: the structured, interpretable rules that capture domain knowledge about AI/ML hiring, and the power of modern semantic embeddings to understand nuanced fit. The hybrid approach ensures both precision (catching role-specific requirements) and recall (finding candidates with non-obvious but relevant backgrounds).